MATHS I · WEEK 2

Maths I week 2 formulas: Coordinate geometry and straight lines

Coordinate geometry · Straight lines · Straight-line fit. Save this PDF for revision.

COORDINATE GEOMETRY · OFTEN ASKED

Coordinate geometry

Distance

$$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$$

Midpoint

$$((x_1 + x_2)/2, (y_1 + y_2)/2)$$

Section, internal m : n

$$((mx_2 + nx_1)/(m + n), (my_2 + ny_1)/(m + n))$$

Section, external m : n

$$((mx_2 - nx_1)/(m - n), (my_2 - ny_1)/(m - n))$$

COORDINATE GEOMETRY · OFTEN ASKED

Coordinate geometry

Centroid

$$((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$$

Area of triangle

$$\tfrac{1}{2} \left|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right|$$

Collinear points

area = 0 (or equal slopes)

Distance from the axes

from x-axis: |y| · from y-axis: |x|

COORDINATE GEOMETRY · OFTEN ASKED

Reflections and quadrants

Reflect (x, y) in	Result
x-axis	(x, −y)
y-axis	(−x, y)
Origin	(−x, −y)
Line y = x	(y, x)
Quadrant signs	I (+, +) · II (−, +) · III (−, −) · IV (+, −)

COORDINATE GEOMETRY · OFTEN ASKED

Common traps

⚠ In the section formula, m multiplies the second point (x_2, y_2).

⚠ Area can't be negative: take the absolute value.

STRAIGHT LINES · IN EVERY PAPER

Straight lines

Slope from 2 points

$$m = (y_2 - y_1)/(x_2 - x_1) = \tan\theta$$

Point–slope

$$y - y_1 = m(x - x_1)$$

Slope–intercept

$$y = mx + c$$

Intercept form

$$x/a + y/b = 1$$

x-intercept a, y-intercept b

STRAIGHT LINES · IN EVERY PAPER

Straight lines

General form

$ax + by + c = 0 \Rightarrow$ slope $-a/b$, x-int $-c/a$, y-int $-c/b$

Horizontal / vertical

$y = k$ (slope 0) · $x = k$ (slope undefined)

Parallel

$m_1 = m_2$

Line parallel to $ax + by + c = 0$: $ax + by + k = 0$

Perpendicular

$m_1 m_2 = -1$

Line perpendicular to $ax + by + c = 0$: $bx - ay + k = 0$

STRAIGHT LINES · IN EVERY PAPER

Straight lines

Angle between lines

$$\tan\theta = |(m_1 - m_2)/(1 + m_1 m_2)|$$

Point to line

$$d = (|ax_1 + by_1 + c|)/(\sqrt{(a^2 + b^2)})$$

Between parallel lines

$$d = (|c_1 - c_2|)/(\sqrt{(a^2 + b^2)})$$

Foot of perpendicular

$$(x - x_1)/a = (y - y_1)/b = -(ax_1 + by_1 + c)/(a^2 + b^2)$$

STRAIGHT LINES · IN EVERY PAPER

Common traps

⚠ Slope of $ax + by + c = 0$ is $-a/b$. Not a/b, not $-b/a$.

⚠ Intercept form needs 1 on the right: $3x + 4y = 12 \rightarrow x/4 + y/3 = 1$.

⚠ $1 + m_1m_2 = 0$ means the lines are perpendicular ($\theta = 90°$), so $\tan\theta$ is undefined.

⚠ Before using the parallel-line distance, make the x and y coefficients identical: $x + 2y - 3 = 0$ becomes $2x + 4y - 6 = 0$ to compare with $2x + 4y + 1 = 0$.

STRAIGHT-LINE FIT · SOMETIMES ASKED

Straight-line fit

Error (residual)

$$e_i = y_i - \hat{y}_i$$

actual – predicted

Sum of squared errors

$$SSE = \Sigma (y_i - \hat{y}_i)^2$$

Better fit

the line with the smaller SSE

Least-squares line

$$m = (n\Sigma xy - \Sigma x\Sigma y)/(n\Sigma x^2 - (\Sigma x)^2),\ c = \bar{y} - m\bar{x}$$

STRAIGHT-LINE FIT · SOMETIMES ASKED

Common traps

⚠ Tip: The least-squares line always passes through $(\bar{x}, \bar{y})$. Use it to knock out options without computing m.

Notes for every week of Maths I, free.

Full notes, the formula sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/maths-1 →

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